SS1

Capstone Project

CAP182

Student: 26445671

PARTA

STADIO Equities Literature Review

Project focus: Predicting which registered accounts are most likely to make a first deposit/activation within 30 days. The three studies below address closely related financial-customer response/subscription prediction problems.

Publication 1

Title	UTAUT-Based Mobile Financial Services Churn Prediction
Harvard reference	Das, A. (2026). *UTAUT-Based Mobile Financial Services Churn Prediction Dataset*. IEEE DataPort. doi:10.21227/t195-ej66.
Link	https://ieee-dataport.org/documents/utaut-based-mobile-financial-services-churn-prediction-dataset
Modelling	The dataset is structured for machine learning and statistical analysis, and participants may apply methods such as classification, regression, clustering, or explainable AI techniques. Common tools include Python (e.g., Scikit-learn, TensorFlow, PyTorch), R, MATLAB, or any equivalent analytical platform.
Preprocessing techniques	Users should preprocess the data appropriately, including handling categorical variables, normalizing Likert-scale responses if required, and defining churn based on inactivity thresholds (e.g., 30+ days)
Datasets Used	We collected the data from a survey conducted from December to March 2026 using a mixed data collection approach, including online platforms, social media, and fieldwork in remote areas. The initial release comprised **937 responses**, which was subsequently expanded to a total of **2,138 responses**, improving coverage and analytical robustness. The dataset includes **24 variables** spanning demographic characteristics (e.g., residence, literacy, income), transactional behavior (e.g., usage frequency and inactivity duration), and psychometric constructs based on the Unified Theory of Acceptance and Use of Technology (UTAUT),

	including Performance Expectancy, Effort Expectancy, Facilitating Conditions, and Perceived Risk.
Key takeaways	Ethical use of data must be maintained at all times. Proper citation of the dataset is required in all derived works Participants are encouraged to report performance metrics such as accuracy, precision, recall, F1-score, and AUC.

Publication2

Title	Machine Learning-Based Prediction of Telecom Customer Churn: Comparative Model Analysis
Harvard reference	Zhang, Y. (2025) 'Machine Learning-Based Prediction of Telecom Customer Churn: Comparative Model Analysis', *Academic Journal of Science and Technology*, 18(2), pp. 116–123. doi: 10.54097/9r12cm81.
Link	https://www.researchgate.net/publication/398104166_Machine_Learning-Based_Prediction_of_Telecom_Customer_Churn_Comparative_Model_Analysis
Modelling	The study compared three classification models**:** Logistic Regression, Random Forest and XGBoost. Logistic Regression was used as the baseline model. Random Forest was used as a bagging ensemble method based on multiple decision trees and bootstrap sampling. XGBoost was used as a gradient-boosting ensemble method that builds trees sequentially to correct previous prediction errors. XGBoost also incorporates regularization to reduce overfitting. We evaluated the models using accuracy, precision, recall, F1-score, and ROC-AUC. XGBoost achieved the highest overall performance, with an AUC of 0.857**.**
Preprocessing techniques	The TotalCharges variable was converted from an object/string type to numeric. Empty values were converted to missing values, and the 11 missing TotalCharges values were replaced with 0, because these records had zero tenure. Binary categorical variables such as Yes/No were encoded as 1/0**.** Categorical variables with three or more categories, such as Contract, Internet Service, and Payment Method, were transformed using one-hot encoding. The study standardized continuous numerical variables, including tenure, Monthly Charges, and Total Charges, using StandardScaler. The dataset was then divided into 80% training and 20% testing data.
Datasets Used	The study used the Telco Customer Churn dataset, originally provided by IBM Analytics. It contains 7,043 customer records and 21 features**,** with Churn as the binary target variable. The dataset contains customer demographic information, account details, service information, and billing-

	related variables. The data had an imbalanced target, with approximately 73.5% non-churn customers and 26.5% churn customers
Key takeaways	**1. Compare multiple models:** The study demonstrates the value of comparing a traditional statistical model with ensemble machine-learning models. Logistic Regression provided a useful baseline, while Random Forest and XGBoost provided stronger predictive performance. **2. Handle categorical and numerical variables appropriately:** The use of binary encoding, one-hot encoding, and standardization provides a useful preprocessing approach for customer churn data. **3. Use appropriate evaluation metrics for imbalanced data:** Accuracy alone can be misleading when churn classes are imbalanced. The study therefore used precision, recall, F1-score, and ROC-AUC alongside accuracy. **4. Feature importance can improve interpretation:** The XGBoost analysis showed that contract type, internet service, tenure, payment method, and charges were important predictors of churn, demonstrating how feature importance can provide additional insight beyond prediction.

Publication 3

Title	Machine Learning Methods for Bank Term Deposit Subscription Prediction
Harvard reference	Xiaoqian Wu (2026). Machine Learning Methods for Bank Term Deposit Subscription Prediction. *International Journal of Global Economics and Management*, *10*(4), 36-43. https://doi.org/10.62051/ijgem.v10n4.06
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Modelling	Compared Logistic Regression, Random Forest, and LightGBM. The study reports LightGBM as the strongest of the three, with 91.95% accuracy and a 60.22% test F1-score. Feature influence was also examined, with call duration and campaign frequency identified as important predictors.
Preprocessing techniques	Missing-value inspection; categorical-variable encoding; relevant feature selection; class-imbalance handling; and data standardization. The study used a common processing framework before comparing the three classifiers.
Datasets Used	Public Bank Marketing dataset. The binary target is whether a customer subscribed to a bank term-deposit product. The study describes customer, marketing, and economic-background variables.
Key takeaways	1. Logistic Regression provides a useful interpretable baseline, while tree ensembles can capture nonlinear relationships.

	2. Class imbalance should be addressed explicitly, and F1 should be considered alongside accuracy. 3. Feature importance can help identify behavioral or campaign characteristics associated with customer response.

UCI Bank Marketing dataset is the one that have public available dataset among the sources reviewed, and its classification target is a customer deposit-subscription response. It can therefore be used as a benchmark/proxy for the STADIOEquities activation problem, while the report should clearly state that term-deposit subscription following a marketing contact is not the same outcome as making a first deposit after investment-account registration. Hence, it will be used for **Part B.**

Name	Bank Marketing
Harvard Reference	Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', *Decision Support Systems*, 62, pp. 22–31. doi:10.1016/j.dss.2014.03.001
Link	https://archive.ics.uci.edu/dataset/222/bank+marketing
Number of instances	45,211 instances for the 17-feature version (bank-full.csv). The repository also provides bank.csv with 4,521 instances (10% sample), plus bank-additional-full.csv with 41,188 instances and 20 input variables, and bank-additional.csv with 4,119 instances.
Number of feature columns	17 input features in the selected bank-full.csv version. The target variable is y, giving 18 columns in total. For a realistic predictive model, the UCI repository explicitly warns that the duration feature should be discarded because it is only known after the contact and is strongly tied to the outcome
Number of classes(if applicable)	Two(2) classes: **yes** and **no**.
Target range(if applicable)	Not applicable because this is a classification dataset. The target y is binary: yes/no.
Origin	The data is related to direct marketing campaigns of a Portuguese banking institution. The marketing campaigns were based on phone calls. Often, more than one contact with the same client was required to determine whether the product (bank term deposit) would be ('yes') or not ('no') subscribed. The UCI Bank Marketing dataset is a real-world banking dataset; it's not synthetically generated.

References

1. Das, A. (2026). *UTAUT-Based Mobile Financial Services Churn Prediction Dataset*. IEEE DataPort. doi:10.21227/t195-ej66.
2. Zhang, Y. (2025) 'Machine Learning-Based Prediction of Telecom Customer Churn: Comparative Model Analysis', *Academic Journal of Science and Technology*, 18(2), pp. 116–123. doi: 10.54097/9r12cm81.
3. Xiaoqian Wu (2026). Machine Learning Methods for Bank Term Deposit Subscription Prediction. *International Journal of Global Economics and Management*, *10*(4), 36-43. https://doi.org/10.62051/ijgem.v10n4.06
4. Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', *Decision Support Systems*, 62, pp. 22–31. doi:10.1016/j.dss.2014.03.001